

Project Report: Customer Behavior Analysis

End-to-End Data Pipeline: Python | SQL | Power BI

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1. Executive Summary

This report outlines a comprehensive analysis of retail customer behavior aimed at enhancing revenue and customer retention. By leveraging an integrated technology stack, the project successfully transformed raw transaction data into a series of actionable business insights. Key findings indicate a total revenue of **\$965.81K** driven significantly by the Electronics category and the 51+ age demographic.

2. Business Problem Definition

The primary objective was to address critical retail challenges including low customer retention and lack of segment-specific marketing strategies. The analysis sought to answer:

- Which categories and products are primary revenue drivers?
- Do discounts positively impact total purchase value or erode margins?
- How do customer demographics (age, gender) influence spending patterns?
- Is there a significant difference in spending behavior between subscribers and non-subscribers?

3. Data Transformation (Python)

Data cleaning was performed using the **Pandas** library to ensure high data integrity.

Key Actions Taken:

- **Standardization:** Mapped specific items to broader categories (e.g., T-shirts to Clothing) to eliminate mixed entries.
- **Imputation:** Handled null values for Size based on category logic (e.g., "Not Applicable" for Electronics). Used median values for Review Ratings and Purchase Amounts to prevent statistical skew.
- **Deduplication:** Identified and removed 50 duplicate records based on Customer ID.
- **Database Export:** Standardized column names for SQL compatibility and exported data to MySQL using SQLAlchemy.

4. Exploratory Data Analysis (SQL)

In-depth querying was utilized to extract granular behavioral insights from the cleaned dataset.

Revenue Insight: Electronics emerged as the top revenue contributor (\$485,769.23), followed by Accessories (\$197,477.29).

Segment Analysis Highlights:

- **Customer Loyalty:** Segmented users into New (0 purchases), Returning (1-10), and Loyal (>10), revealing a massive loyal base of 3,537 customers.
- **Discount Efficiency:** Identified "Premium Discount Users"—customers who utilize discounts but still spend above the average transaction value.
- **Product Performance:** Ranked the top 3 most purchased items per category using SQL Window Functions.

5. Data Visualization & Insights (Power BI)

An interactive dashboard was developed to facilitate executive decision-making.

- **Demographics:** The 51+ age group contributes the highest revenue (\$387K), suggesting a need for targeted loyalty programs for older demographics.
- **Shipping Preferences:** Express shipping users showed a higher average transaction value (\$345.07) compared to Standard shipping.
- **Transaction Trends:** Seasonal analysis shows relatively stable performance across Spring, Winter, and Summer, with a notable dip in Autumn.

6. Conclusion & Recommendations

The project confirms that a data-driven approach to retail can uncover hidden growth opportunities.

Strategic Recommendations:

- **Targeted Loyalty:** Expand marketing efforts toward the 51+ age bracket and the "Loyal" customer segment.
- **Category Optimization:** Increase inventory and marketing spend on high-performing Electronics and Accessories.

- **Premium Services:** Promote Express shipping and Subscription models, as these cohorts show higher Average Order Values (AOV).